

Practice Projects

For practice, write a function for each of the following tasks:

1. Create a version of the `karatsuba()` function that has a multiplication lookup table of products from 0×0 to 999×999 rather than 0×0 to 9×9 . Get a rough estimate of how long it takes to calculate `karatsuba(12345678, 87654321)` 10,000 times in a loop with this larger lookup table compared to the original lookup table. If this still runs too quickly to measure, increase the number of iterations to 100,000 or 1,000,000 or more. (Hint: you should delete or comment out the `print()` and `document.write()` calls inside the `karatsuba()` function for this timing test.)
2. Create a function that performs a linear search on a large array of integers 10,000 times. Get a rough estimate of how long this takes, increasing the number of iterations to 100,000 or 1,000,000 if the program executes too quickly. Compare this with how long a second function takes to sort the array once before performing the same number of binary searches.